

Poll

What is your current role?

1. Data Analyst
2. Data Engineer
3. Data Scientist
4. Non Data Related Role

Poll

What level of experience do you have with Machine Learning?

1. None
2. Novice
3. Advanced Beginner
4. Competent

Poll

What level of experience do you have with **Azure** Machine Learning?

1. None
2. Novice
3. Advanced Beginner
4. Competent



Segment 1: Introduction to Machine Learning and Azure ML

What is Machine Learning?

Software “models” are trained from data to predict certain patterns, trends, and outcomes

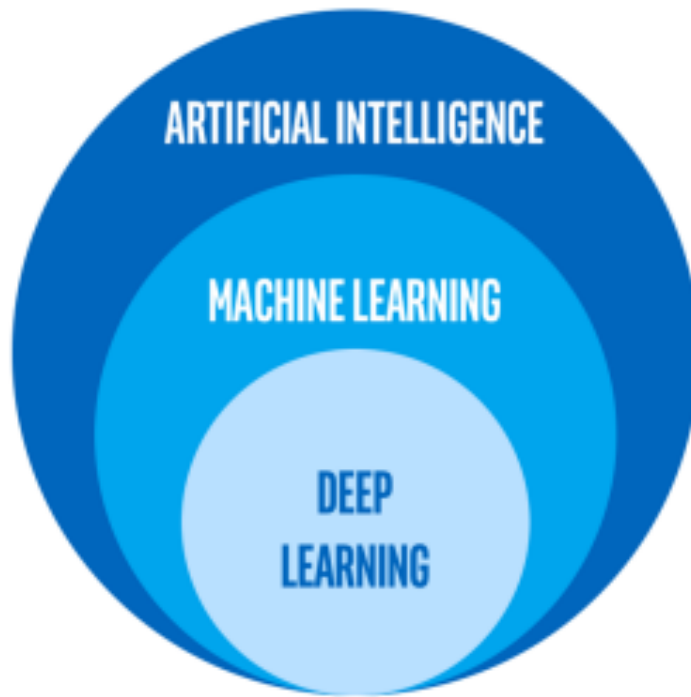
Traditional Programming



Machine Learning

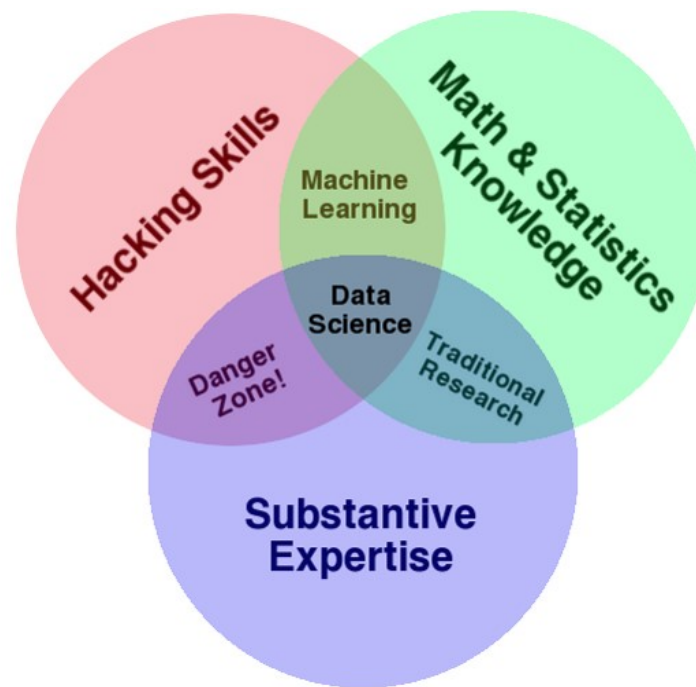


What is Machine Learning?



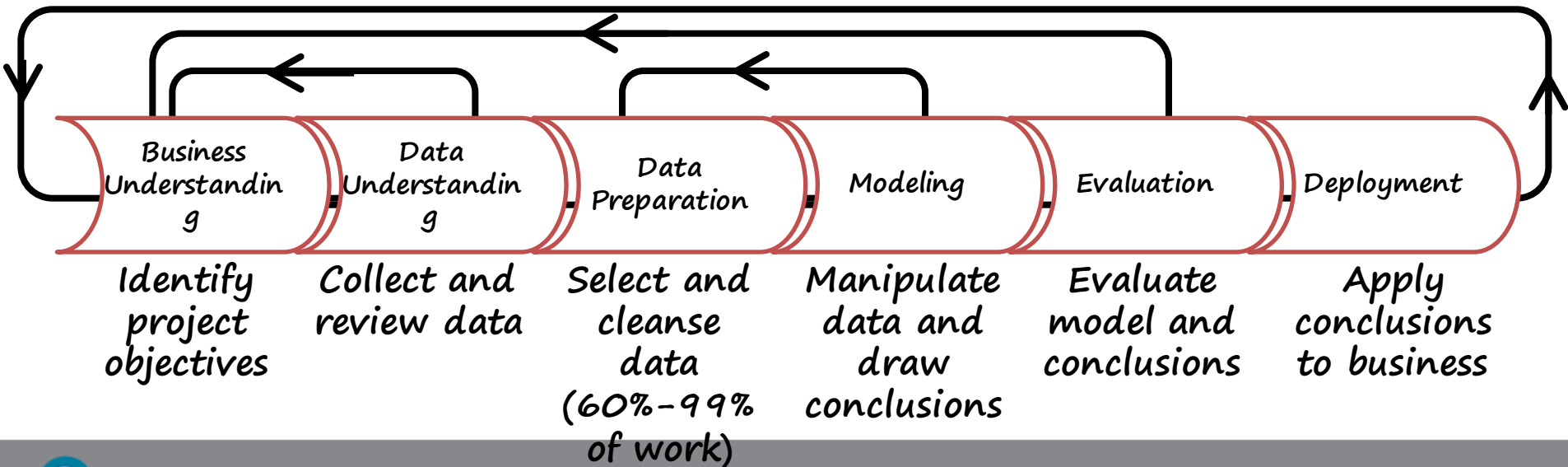
Software “models” are trained from data to predict certain patterns, trends, and outcomes

What is Machine Learning?



What is Machine Learning?

Create "experiments," evaluate, and then "fail fast".



What is Azure Machine Learning?



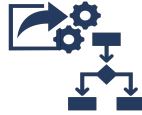
Azure Machine Learning



Scalable, on-demand compute



Data storage and connectivity



ML workflow orchestration



Model registration and management



Metrics and monitoring

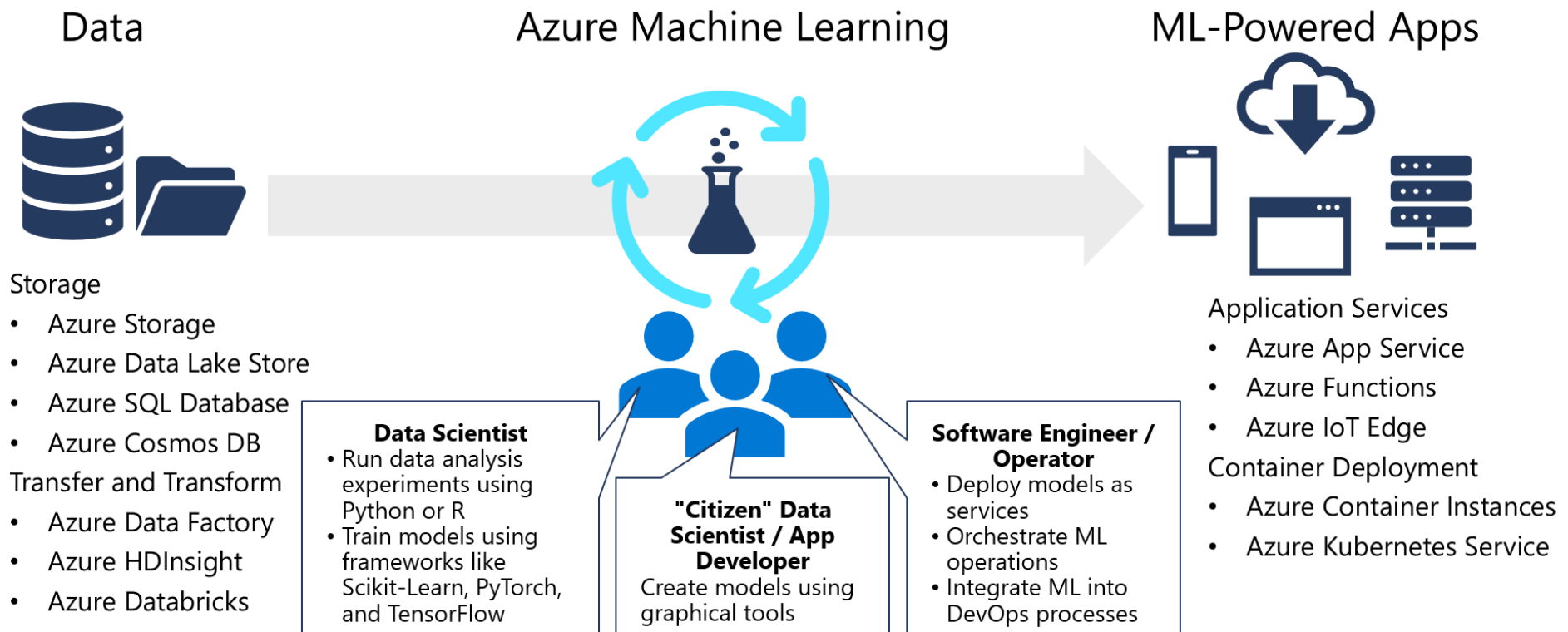


Model deployment



Microsoft Azure

What is Azure Machine Learning?

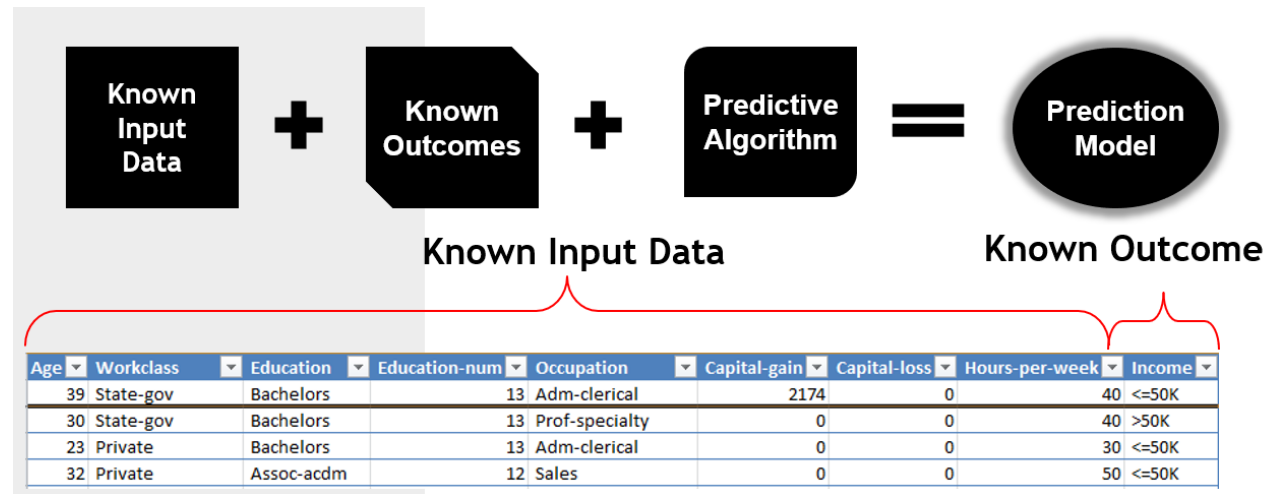


What is Azure Machine Learning?

1. A platform for operating machine learning workloads in the cloud
2. A service to spin up an R Cluster in Azure
3. A tool to analyze algorithms developed on-prem
4. An Azure service to access AI capabilities like Computer Vision

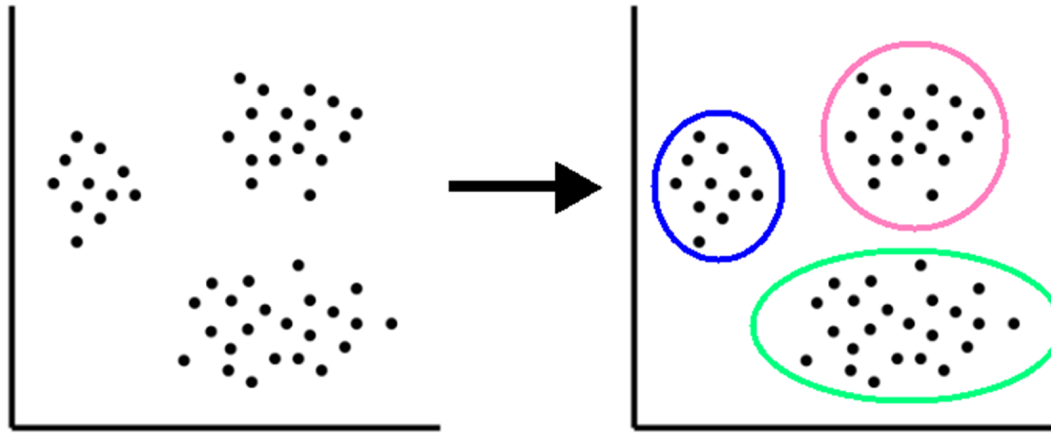
How to select the right algorithms

- Supervised Learning
 - Model based on known input data and outcomes.

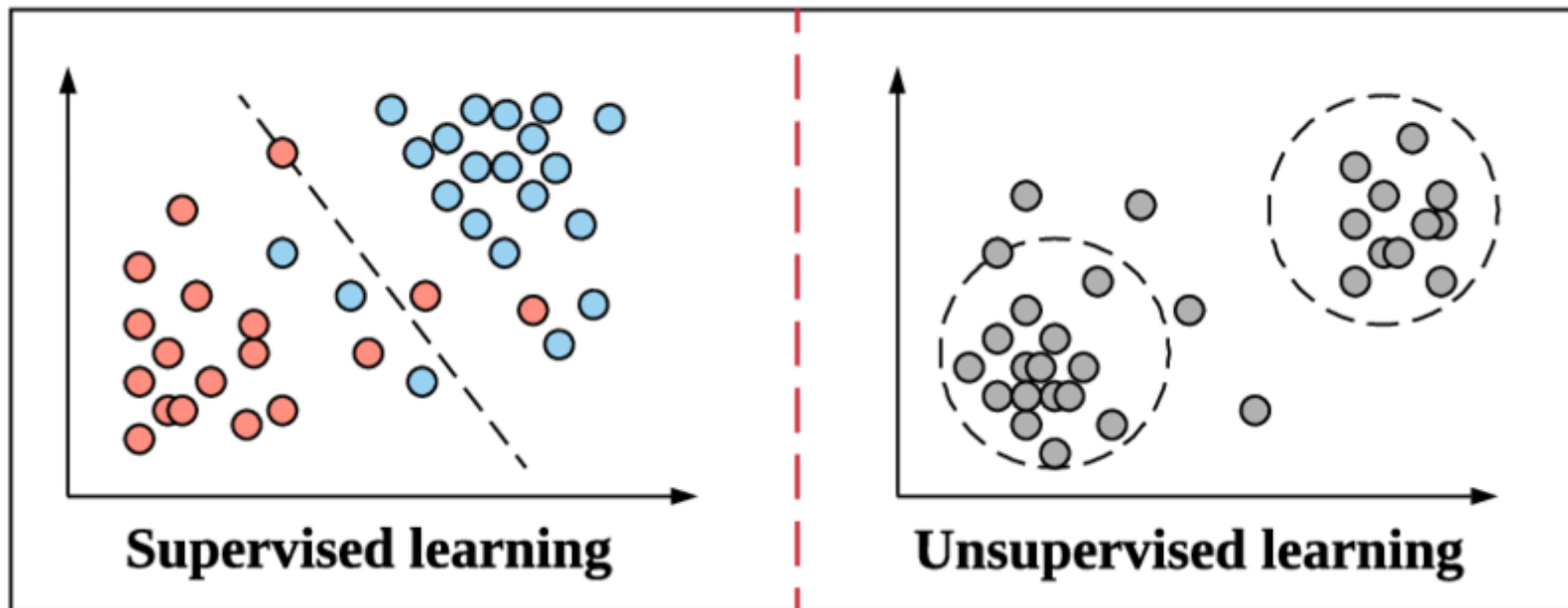


How to select the right algorithms

- Unsupervised Learning: no labels



How to select the right algorithms

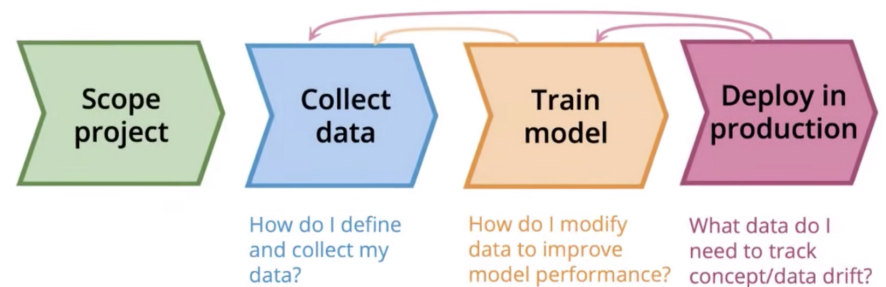
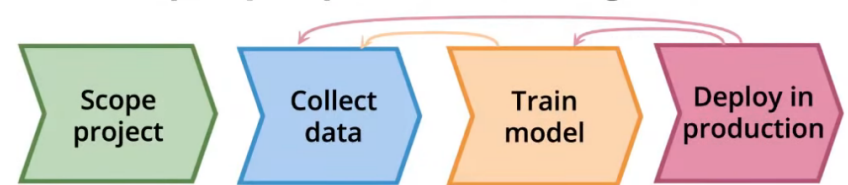


Supervised learning uses ____ data, while unsupervised does not.

1. numerical
2. labeled
3. categorical
4. ordered

Data Centric Machine Learning

- **Model-centric approach:**
Asks how you can change the model to improve performance.
- **Data-centric approach:**
Asks how you can change or improve your data to improve performance.



Demo: Azure Machine Learning

Designer





Break

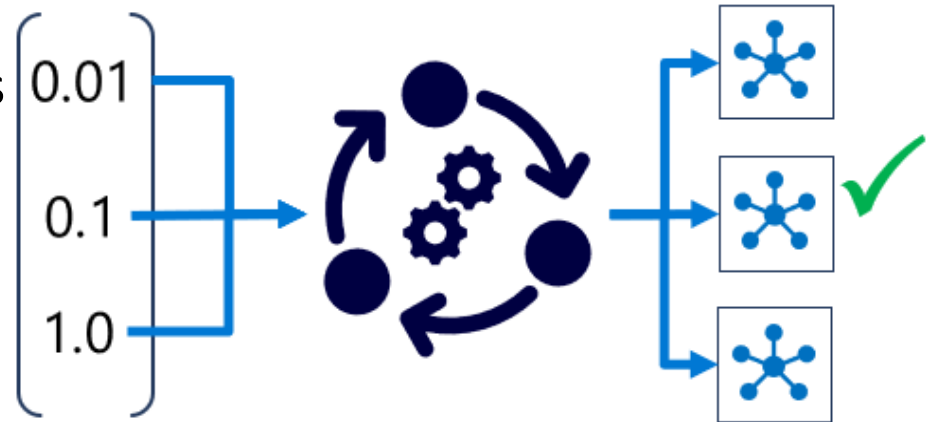




Segment 2: Improve Your Model

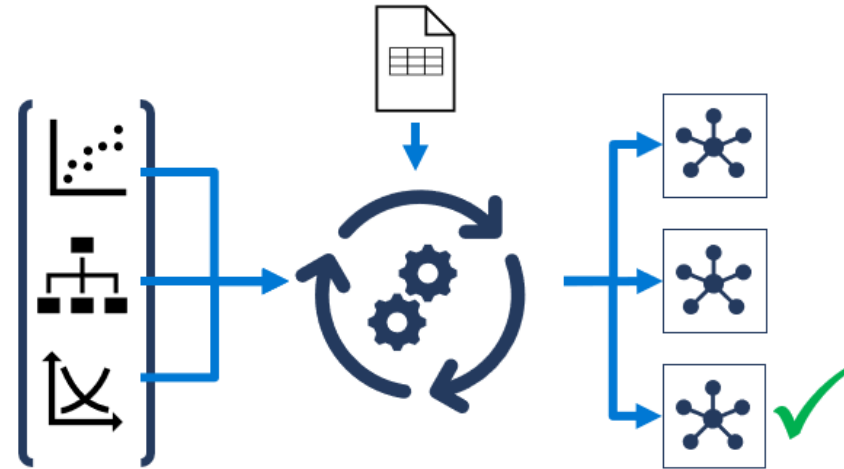
What is hyperparameter tuning?

- Train multiple models, using the same algorithm but varying hyperparameter values
- Find the "best" model based on a specific performance metric



What is Automated Machine Learning?

- Train multiple models in parallel, varying algorithm and preprocessing
- Find the "best" model based on a specific performance metric



What is Automated Machine Learning?

The screenshot displays the 'Start run' page in the Microsoft Azure Machine Learning portal. The browser address bar shows the URL <https://ml.azure.com/automl/startrun?wsid=/subscriptions/>. The page title is 'Microsoft Azure Machine Learning'. The breadcrumb navigation indicates the path: 'gmalc-dp100 > Automated ML > Start run'.

The main heading is 'Create a new Automated ML run'. On the left, a vertical sidebar contains icons for various ML operations. A progress indicator shows three steps: 'Select dataset' (checked), 'Configure run' (checked), and 'Task type and settings' (active).

The 'Task type and settings' section is titled 'Select task type'. It includes the instruction: 'Select the machine learning task type for the experiment. Additional settings are available to fine tune the experiment if needed.'

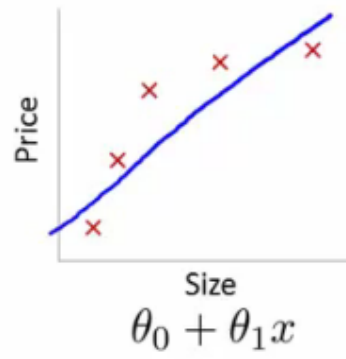
Three task type options are listed:

- Classification**: To predict one of several categories in the target column. yes/no, blue, red, green. This option is selected, indicated by a green checkmark.
- Regression**: To predict continuous numeric values.
- Time series forecasting**: To predict values based on time.

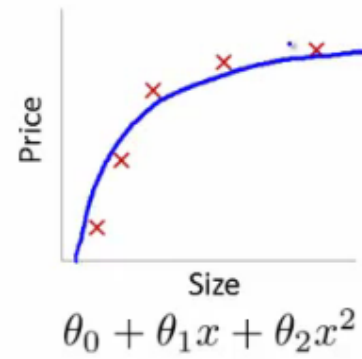
Below the task type options, there is a checkbox for 'Enable deep learning' which is currently unchecked. At the bottom of the settings section, there are two links: 'View additional configuration settings' and 'View featurization settings'.

At the bottom of the page, there are three buttons: 'Back', 'Finish' (highlighted in blue), and 'Cancel'.

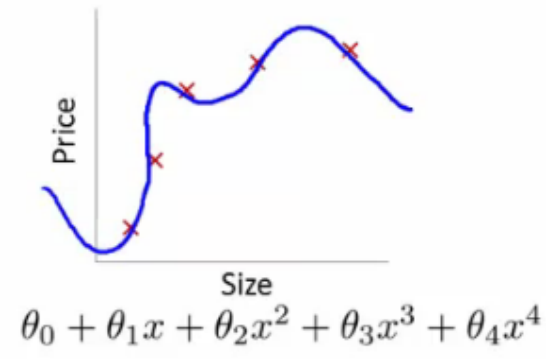
Bias/Variance Tradeoff



High bias
(underfit)

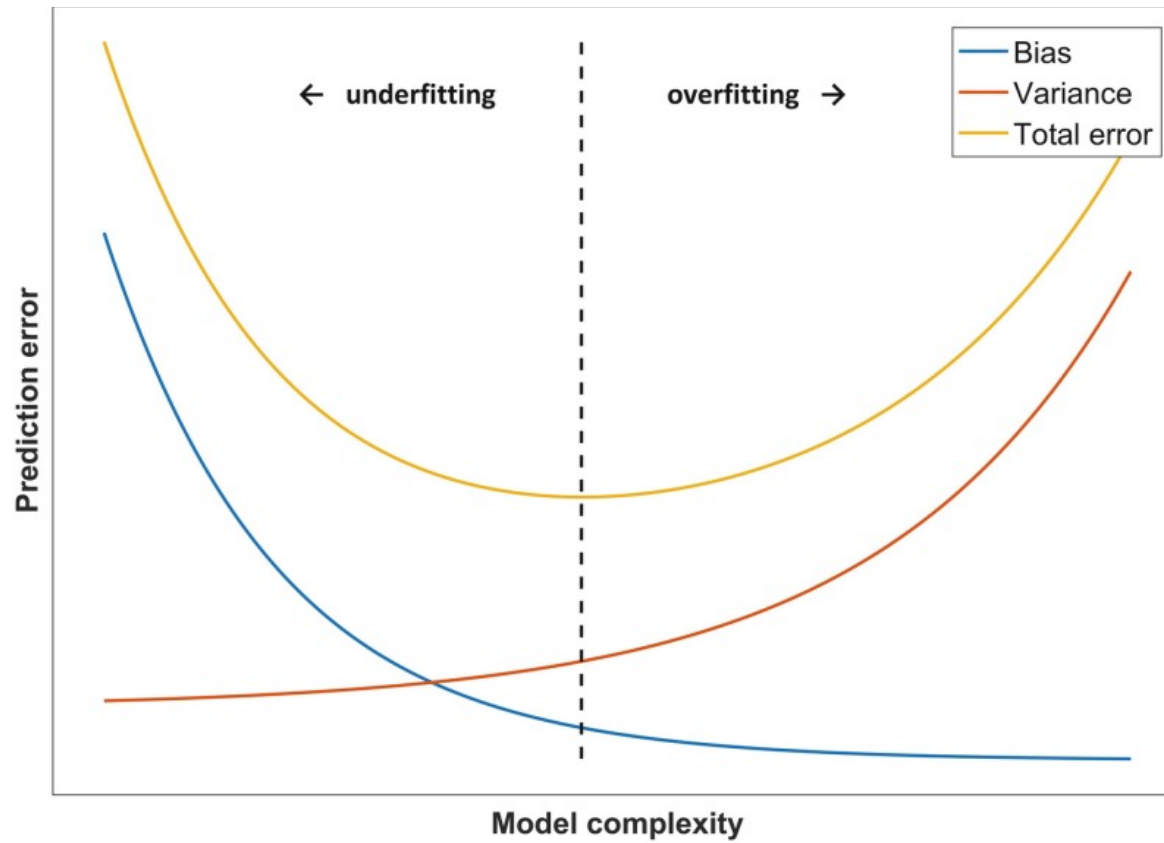


"Just right"



High variance
(overfit)

Bias/Variance Tradeoff



Bias/Variance Tradeoff

- Run 1
 - Train Dataset: 78% Accuracy
 - Test Dataset: 79% Accuracy

Bias/Variance Tradeoff

- Run 1
 - Train Dataset: 78% Accuracy
 - Test Dataset: 79% Accuracy
- Run 2
 - Train Dataset: 85% Accuracy
 - Test Dataset: 84% Accuracy

Bias/Variance Tradeoff

- Run 1
 - Train Dataset: 78% Accuracy
 - Test Dataset: 79% Accuracy
- Run 2
 - Train Dataset: 85% Accuracy
 - Test Dataset: 84% Accuracy
- Run 3
 - Train Dataset: 95% Accuracy
 - Test Dataset: 81% Accuracy

If the performance of my algorithm is much better on the train dataset than on the test dataset, then:

1. The model is low bias
2. The model should be complexified further
3. The model is overfitting
4. The model is low variance

Demo: Automated Machine Learning





Break

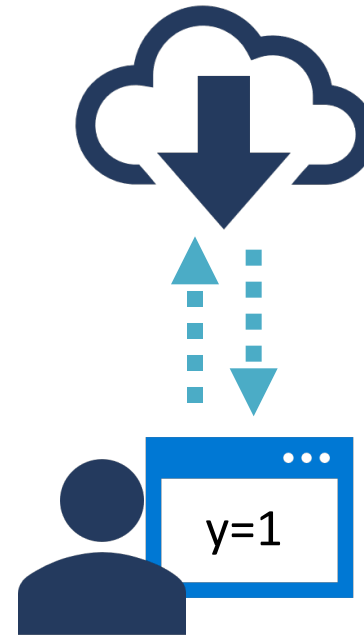




Segment 3: Deploy and Monitor Your Model

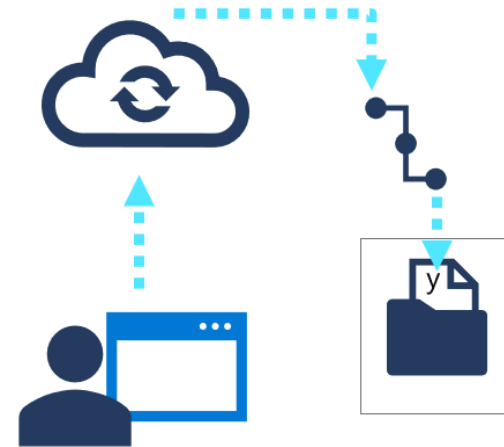
Deploy and consume your model

- Real-Time Inferencing
- Immediate prediction from new data
 - Usually deployed as a web service endpoint



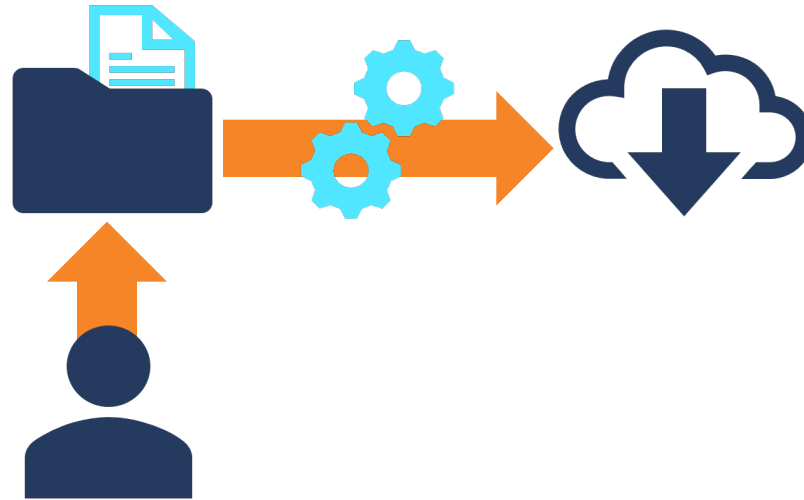
Deploy and consume your model

- Batch Inferencing
- Asynchronous prediction from batched data
- Implemented as a pipeline
 - Typically using a `ParallelRunStep` for scalability



CI/CD with Machine Learning

- A core DevOps practice for software development and deployment
 - Source control system
 - Updates trigger build and release



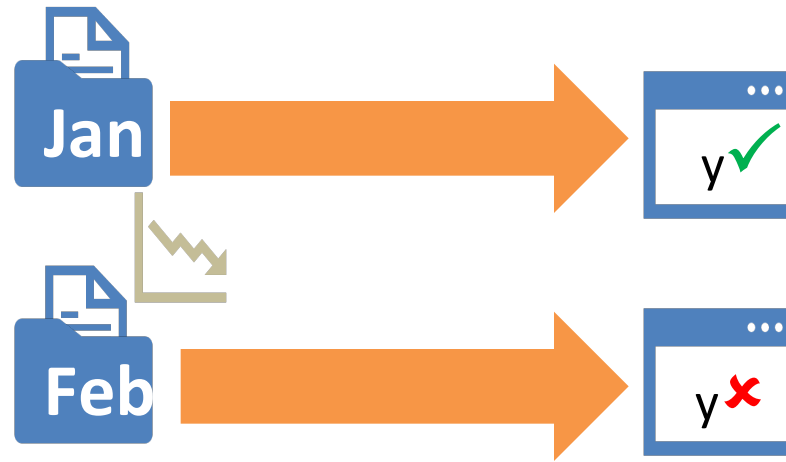
CI/CD with Machine Learning

- Azure Machine Learning and Azure Pipelines
 - Pipelines to train and deploy models
 - Azure Pipelines *Machine Learning* extension



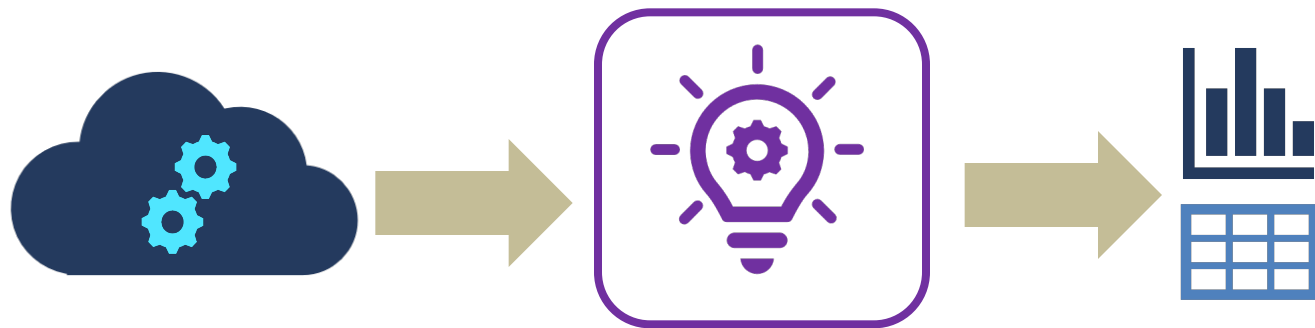
Monitoring Data Drift

- Changing data trends that can affect the accuracy of trained models
- Data Drift Detector
 - Monitor by comparing Datasets



Monitoring with Application Insights

- An Application Performance Management service in Azure
- Enables capture, storage, and analysis of telemetry data



Monitoring data drift helps to know...

1. If the data has new variables to include in the model
2. How much the model is used in production
3. When to retrain the model
4. If the model is under attack by a third party

Demo: Application Insights





Break

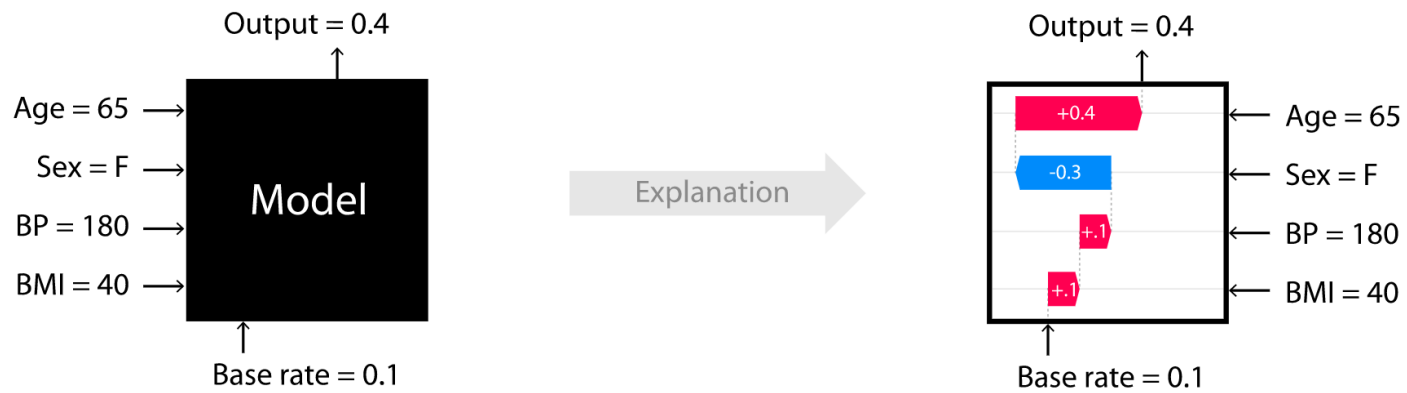




Segment 4: Interpret Your Model

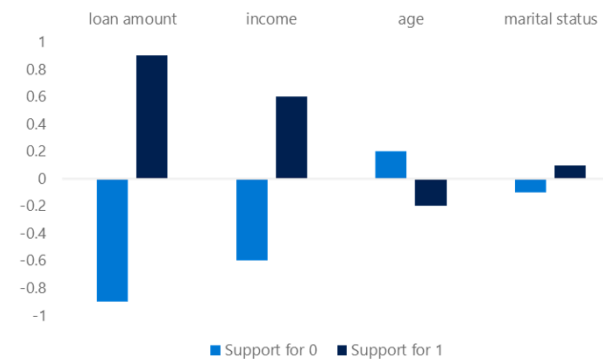
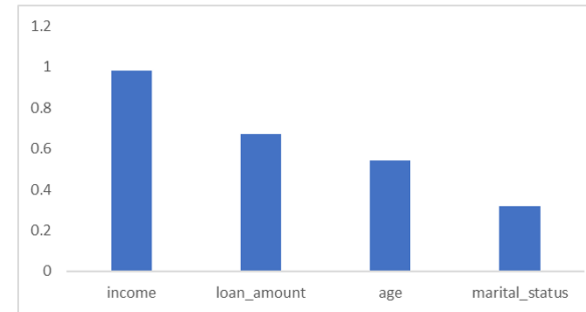
Introduction to Explainers

- Why interpreting is important?



Global and Local feature importance

- **Global Feature Importance**
 - For all test data
 - Relative influence of each feature on the predicted label
- **Local Feature Importance**
 - For an individual prediction
 - Relative support for each possible class per feature



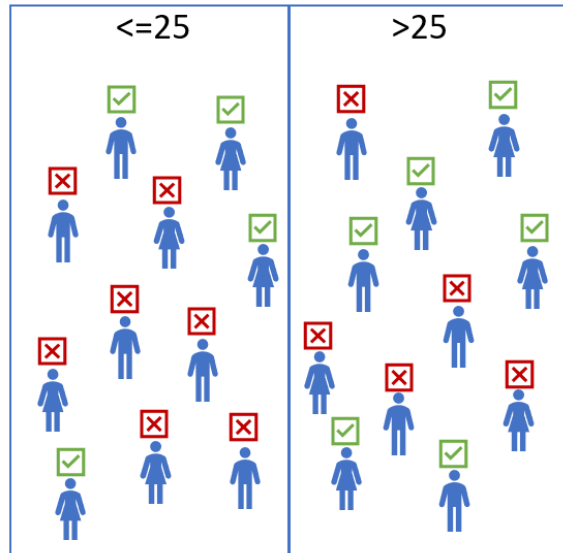
Poll

Explaining a model is only important when:

1. The data was synthesized by a GAN
2. The model will be included in a research paper
3. Decision makers will use the model
4. It's always important

How to detect and mitigate fairness

Selection Rate Disparity



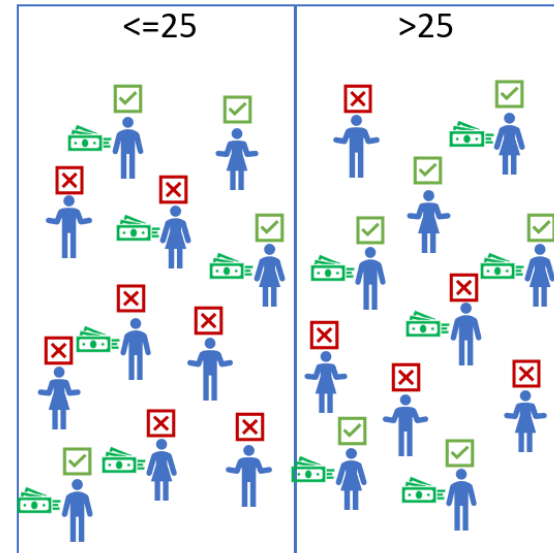
Overall selection rate = $10/22$ (45%)

25 & under selection rate = $4/11$ (36%)

Over 25 selection rate = $6/11$ (54%)

Disparity = 18%

Prediction Performance Disparity



Overall recall = $8/12$ (67%)

25 & under recall = $3/6$ (50%)

Over 25 recall = $5/6$ (83%)

Disparity = 33%

Demo: Interpret Your Model







Course wrap-up and next steps

Wrap-up

- ML and Azure ML
- Improve your model
- Deploy and monitor your model
- Interpret your model

Next Steps

- DP-100 Certification
- Databricks